14 March 1977

SUPERSEDING
MIL-M-55198(EL)
19 October 1962

# MILITARY SPECIFICATION

## MICROPHONE, DYNAMIC M-116( )/G

This specification is approved for use by the Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers one type of dynamic noise cancelling microphone; designated as Microphone, Dynamic M-116( )/G.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| T-T-871 | Twine, Fibrous Cotton |
| NN-P-71 | Pallet, Materials-handling, Wood |
| QQ-S-781 | Strapping, Steel, Flat and Seals |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: CG, US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, N.J. 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5965

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SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-B-566 | Box, Folding, Paperboard |
| PPP-B-585 | Box, Wood, Wirebound |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner |
| PPP-B-636 | Boxes, Fiber |
| PPP-B-640 | Box, Fiberboard, Corrugated, Triple Wall |
| PPP-B-676 | Box, Setup |
| PPP-C-795 | Cushioning material, Plastic Film |
| PPP-C-843 | Cushioning material, Cellulosic |
| PPP-C-1797 | Cushioning material, Polypropylene Foam |
| PPP-F-320 | Fiberboard, Corrugated and Solid Sheet Stock |
| PPP-S-760 | Strapping, Nonmetallic (and Connectors) |
| PPP-T-97 | Tape, Pressure-Sensitive Adhesive |

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation-packaging, Methods of |
| MIL-P-11268 | Parts, Materials, and Processes Used in Electronic Equipment |
| MIL-M-13231 | Marking of Electronic Items |
| MIL-F-14072 | Finishes for Ground Electronic Equipment |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-147 | Palletized and Containerized Unit Loads |
| MIL-STD-202 | Test Methods for Electronic and Electrical Component Parts |
| MIL-STD-252 | Wired Equipment, Classification of Visual and Mechanical Defect |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |
| MIL-STD-810 | Environmental Test Methods |



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DRAWINGS

ELECTRONICS COMMAND

| | |
|---|---|
| SC-DL-208640 | Microphone, Dynamic M-116( )/G |
| SC-D-621083 | Microphone and Earphone Test Chamber |

NAVY

| | |
|---|---|
| SK-N-864 | Simulated Gun Blast Producing Equipments |

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE

S1.10 - Calibration of Microphones
S1.12 - Specifications for Laboratory Standard Microphones

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS

258 - Test Procedure for Close-Talking Pressure-Type Microphones

(Application for copies should be addressed to the IEEE Order Dept., 445 Hoes Lane, Piscataway, N.J. 08854)

3. REQUIREMENTS

3.1 First Article. - When specified in the contract or purchase order, the contractor shall furnish sample units for first article inspection and approval (See 4.3).



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3.2 Construction. - The equipment shall be constructed in accordance with the requirements of this Specification and Drawing and Data List SC-DL-208640 (including the requirements for parts, materials and processes thereon) (See 4.4).

3.3 Parts, materials and processes; general. - In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply (See 4.4).

3.4 Finish. - The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings (See 4.4).

3.5 Marking. - Marking shall conform to Specification MIL-M-13231 (See 4.4).

3.6 Performance characteristics.

3.6.1 Air Stop. - The air stop at the microphone end of the cable shall not allow passage of air when tested in accordance with 4.7.2.1.

3.6.2 Insulation resistance. - The insulation resistance between the connector pins and all external metal parts shall be not less than 10 megohms. (See 4.7.2.2)

3.6.3 Operation. - The equipment shall transmit intelligible voice signals without buzzing, rattling or other spurious sounds. (See 4.7.2.3)

3.6.4 Response. - The minimum power output of the microphone shall be -56 dBm (0.63 millivolts across 150 ohms) with a 1000 Hz sound pressure input of 28 dynes per square centimeter. The frequency response in the range of 300 to 7000 Hz shall fall within the envelope of Figure 1.

3.6.5 Distortion. - Total harmonic distortion shall not exceed 5 percent over the frequency range of 300 to 3500 Hz.

3.6.6 Impedance. - The output impedance of the microphone at 1000 Hz shall be 150 ohms +0 or -20 percent.

3.6.7 Signal to noise ratio. - The ratio of signal to noise shall be not less than 15 dB.



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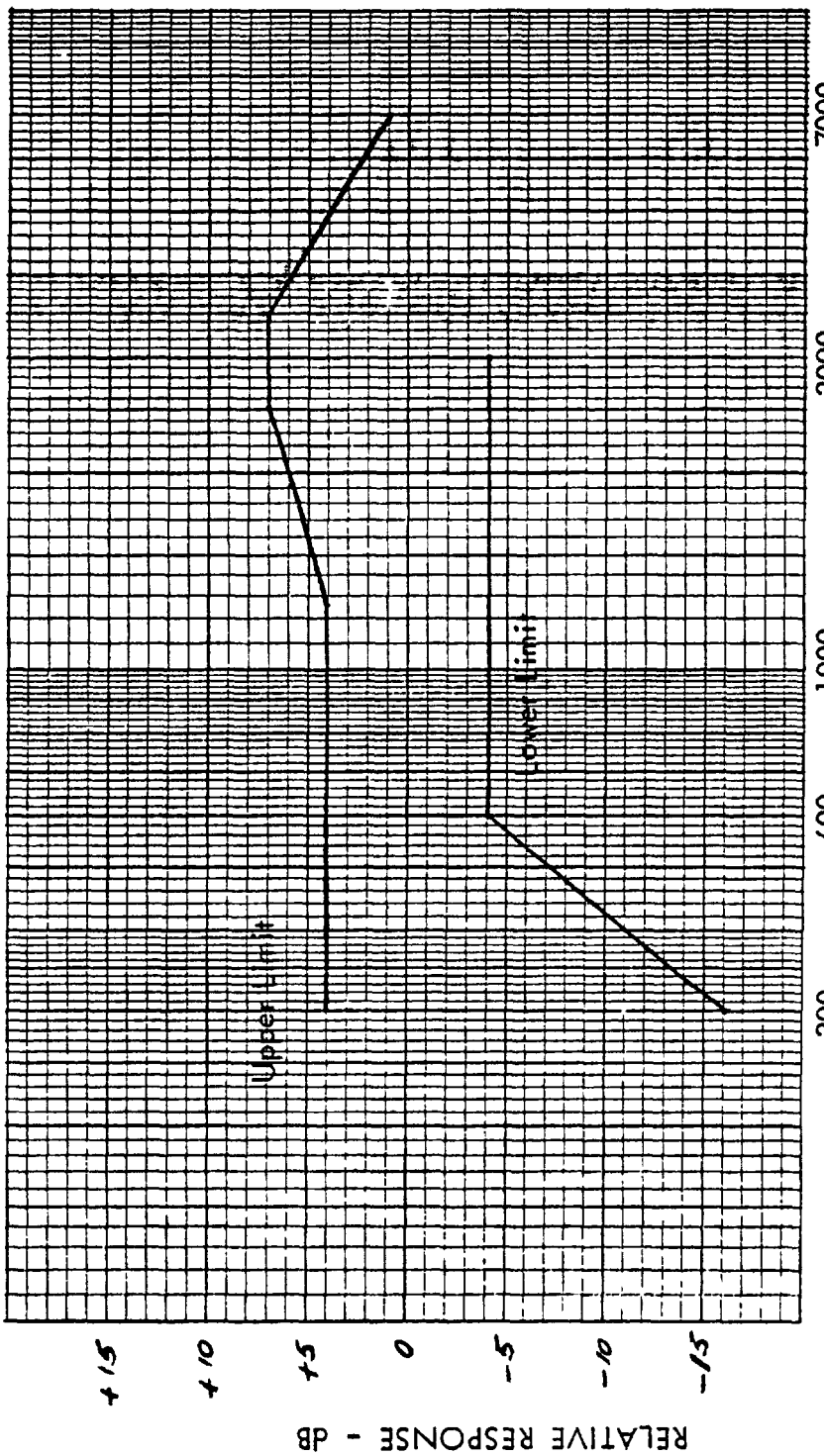



Figure 1 Microphone Response Limit Curves



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3.7 Environmental. - The equipment shall meet the following environmental conditions.

3.7.1 Temperature.

3.7.1.1 Low. - During and after testing as specified in 4.8.1.1, the equipment shall meet the requirements of 3.6.4 with no more than a 3 dB change in response.

3.7.1.2 High. - During and after testing as specified in 4.8.1.2, the equipment shall meet the requirements of 3.6.4 with no more than a 3 dB change in response.

3.7.2 Humidity. - The equipment shall exhibit no physical damage such as corrosion, rust, blistering, swelling or deterioration of parts and materials, and shall meet the requirements of 3.6.4 during the operating times specified in 4.8.2.

3.7.3 Altitude (Elevation). - The equipment shall meet the requirements of 3.6.4 with no more than a 5 dB change in frequency response at altitudes up to 10,000 feet above sea level after storage at 50,000 feet above sea level. (See 4.8.3)

3.7.4 Leakage (Immersion). - The equipment shall meet the requirements of 3.6.2 and 3.6.4 following the test of 4.8.4

3.7.5 Blast. - The equipment shall meet the requirements of 3.6.4 with no more than a 3 dB change in frequency response following thirty rounds of blast at a peak pressure of 9.5 pounds per square inch (psi). (See 4.8.5)

3.7.6 Shock, drop. - The equipment shall meet the requirements of 3.6.4 with no more than a 3 dB change in frequency response and there shall be no evidence of physical damage following the tests of 4.8.6.

3.7.7 Bounce, loose cargo. - The equipment shall meet the requirements of 3.6.4 following the test of 4.8.7.

3.7.8 Salt fog. - After the salt fog test of 4.8.8, the equipment, when examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surface, or exfoliation of metal. Electrical operation is not required.



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3.8 Interchangeability. - Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 (See 4.9).

3.9 Workmanship. - The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 (See 4.6).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. - The inspection requirements specified herein are classified as follows:

a. First Article inspection (See 4.3). Does not include preparation for delivery.

b. Inspections covered by subsidiary documents (See 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery (See 4.5).

(2) Quality conformance inspection of preparation for delivery (See 4.10).

4.3 First Article. - Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor.

4.3.1 First Article units. - The contractor shall furnish nine First Article units of Microphone, Dynamic M-116( )/G.



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4.3.2 First Article inspection. - The First Article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, and the inspections specified for Group A, Group B and Group C (See Tables I,II,III respectively). The inspection shall be performed in the following order: 4.4, Group A and Group B for all units; and Group C as specified in Table III. After completion of Group C environmental tests, conforming units shall be reinspected and shall pass all Group A inspection.

4.3.3 First Article data. - The First Article test plan and test report shall be furnished as required in the contract or purchase order.

4.3.4 Failures. - One or more failures shall be cause for refusal to grant first article approval.

4.4 Inspections covered by subsidiary documents: The following shall be inspected under the applicable subsidiary document as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

| Item | Where required |
|---|---|
| Construction | 3.2 |
| Parts, materials and processes-general | 3.3 |
| Finish | 3.4 |
| Marking | 3.5 |

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4, 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. - Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQLs indicated in Table I.



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4.5.1.1 Order of inspection within Group A. - Group A inspection shall be performed in an order satisfactory to the Government except that the operational inspection shall be last.

Table I. Group A inspection

| Inspection | Req Para | Insp Para | AQL | |
|---|---|---|---|---|
| | | | Major | Minor |
| Visual & Mechanical | 3.9 | 4.6 | 1% | 4% |
| Air Stop | 3.6.1 | 4.7.2.1 | 1% | * |
| Electrical | | | | |
| Ins Resistance | 3.6.2 | 4.7.2.2 | 1% | * |
| Operation | 3.6.3 | 4.7.2.3 | 1% | * |
| Response | 3.6.4 | 4.7.2.4 | 1% | * |
| Impedance | 3.6.6 | 4.7.2.6 | 1% | * |

* All defects are considered major.

4.5.2 Group B Inspection. - Group P inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table II and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B Sampling Plans. - The Group B sampling plans for the AQLs listed in Table II shall be Inspection Level S-4.

4.5.2.2 Order of Inspection Within Group B. - Group P inspection shall be performed in an order satisfactory to the government.

Table II Group B inspection

| Inspection | Req Para | Insp Para | AQL |
|---|---|---|---|
| Interchangeability | 3.8 | 4.9 | 6.5% |
| Distortion | 3.6.5 | 4.7.2.5 | 4.0% |



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4.5.3 Group C Inspection. - Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table III. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection. - For Subgroup 1, two samples each of the Microphone shall be selected at random from each 500 units or fraction thereof produced. Each sample shall be tested to all tests of the Subgroup. For Subgroup II, five samples shall be selected at random from each 1000 units or faraction thereof produced. One sample shall be tested against each test. For Subgroup III, two samples shall be selected at random from each 2000 units or fraction thereof produced. One sample shall be tested against each test. The first samples of each test subgroup shall be selected from the first production lot.

4.5.3.2 Group C Failures. - Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of Conforming Group C Sample Units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass Group A and Group B inspection.

Table III Group C inspection

| Inspection | Req Para | Insp Para |
|---|---|---|
| Subgroup I | | |
| Signal to Noise | 3.6.7 | 4.7.2.' |
| Leakage(Immersion) | 3.7.4 | 4.8.4 |
| Subgroup II | | |
| Temperature | 3.7.1 | 4.8.1 |
| Humidity | 3.7.2 | 4.8.2 |
| Altitude | 3.7.3 | 4.8.3 |
| Shock, drop | 3.7.6 | 4.8.6 |
| Bounce | 3.7.7 | 4.8.7 |
| Subgroup III | | |
| Blast | 3.7.5 | 4.8.5 |
| Salt fog | 3.7.8 | 4.8.8 |



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4.6 Visual and mechanical. - The equipment shall be examined for compliance with 3.9. Classification of defects shall be in accordance with MIL-STD-252.

4.7 Performance requirements.

4.7.1 Test equipment. -

a. Vacuum tube voltmeter (VTVM). - Ballantine model 300 VTVM or equal.

b. Sound source. - Artificial Voice in accordance with IEEE Standard 258.

c. Audio oscillator. - General Radio Company type 1304-B or equal.

d. Standard microphone. - Shall be a Laboratory-Type microphone in accordance with ANSI S1.12, calibrated by the reciprocity method in accordance with ANSI S1.10.

e. Sound pressure calibration. - Mount the standard microphone concentrically and flush in a baffle having the same shape as the cap of the microphone housing. Mount the baffle and standard microphone coaxially and ½ inch away from the sound source. Adjust and record the audio oscillator input to the sound source to obtain 28 dynes per square centimeter (as measured at the standard microphone output) at the frequencies of 300, 400, 500, 600, 800, 1000, 1200, 1500, 2000, 2500, 3000, 3500, 4000, 5000, 6000 and 7000 Hz as a minimum.

f. Distortion. - Hewlett Packard Distortion Analyzer model 330C or equal.

4.7.2 Methods of inspection.

4.7.2.1 Air Stop. - Immerse the microphone end of the cable in water to a depth that will cover the air stop portion. Apply four pounds of air pressure to the connector end of the cable for ten minutes. Determine compliance with 3.6.1.

4.7.2.2 Insulation resistance. - The assembly shall be tested in accordance with Method 302, test condition B, of MIL-STD-202 to meet the requirements of 3.6.2. There shall be no evidence of arcing or insulation breakdown during the test.



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4.7.2.3 Operation. - Connect the microphone under test through a suitable audio amplifier, having essentially flat response in the audio frequency range, to a speaker. Determine compliance with 3.6.3.

4.7.2.4 Response. - The microphone under test, in its housing, shall be mounted coaxially and ¼ inch from the calibrated sound source. Terminate the microphone under test with a 150 ohm non-inductive resistor. Apply a constant sound pressure level of 28 dynes per square centimeter at the frequencies cited in 4.7.1.e and measure the output voltage with a VTVM to determine compliance with 3.6.4,

4.7.2.5 Distortion. - Repeat the tests of 4.7.2.4 except that the constant sound pressure level shall be 125 dB above a reference level of .0002 dynes per square centimeter. Measure total harmonic distortion for compliance with 3.6.5.

4.7.2.6 Impedance. - Repeat the test of 4.7.2.4 except that the sound pressure level at 1000 Hz shall be adjusted to that level which produces .001 volts at the open circuit output terminals of the microphone under test. Holding the sound pressure level constant, connect a decade resistance box to the microphone terminals and adjust its value until the output voltage is .0005 volts. The decade resistance setting shall comply with 3.6.6. Repeat test of frequencies of 300 and 3500 Hz.

4.7.2.7 Signal to noise ratio. -

a. Tests shall be performed in a suitable acoustical environment similar to that shown in Drawing SC-D-621083.

b. The Noise source shall consist of nine separately generated audio frequencies as shown in Table IV with their relative sound pressure levels. The Noise source shall be located as far away from the microphone as is practicable and on a line coincident with the central plane of the microphone. By means of a calibrated standard microphone, adjust the Noise output for a sound pressure of 115 dB above 0.0002 dynes per square centimeter as measured at the standard microphone.



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c. The Signal source shall consist of seven separately generated audio frequencies as shown in Table V with their relative sound pressure levels. The Signal source shall be adjusted to supply a sound pressure level of 115 dB above 0.0002 dynes per square centimeter to a standard microphone positioned coaxially one quarter inch away from the sound source.

d. With the microphone under test positioned in place of the standard microphone (one quarter inch from the Signal source) activate the Signal source and measure the output voltage across a 150 ohm non-inductive resistor with a VTVM. Shut off the Signal source, activate the Noise source and measure the microphone output voltage. The difference in the microphone output with Signal input and the microphone output with Noise input, in dB, shall meet the requirement of 3.6.7.

Table IV Noise Spectrum

| Frequency in Hz | Sound Pressure in dB Relative to Sound Pressure at 130 Hz |
|---|---|
| 40 | 0 |
| 70 | 0 |
| 130 | 0 |
| 300 | -5 |
| 600 | -9 |
| 1000 | -13 |
| 2000 | -17 |
| 3000 | -19 |
| 4000 | -21 |

Table V Signal Spectrum

| Frequency in Hz | Sound Pressure in dB Relative to Sound Pressure at 130 Hz |
|---|---|
| 130 | 0 |
| 300 | +7 |
| 600 | +8 |
| 1000 | +5 |
| 2000 | -3 |
| 3000 | -7 |
| 4000 | -9 |



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4.8 Environmental conditions.

4.8.1 Temperature.

4.8.1.1 Low. - The equipment shall be tested in accordance with Method 502.1, Procedure 1 of MIL-STD-810. The storage temperature shall be minus 62°C, maintained for four hours, and the operating temperature shall be minus 40°C. Prior to, during and following the test, the equipment shall meet the requirements of 3.7.1.1.

4.8.1.2 High. - The equipment shall be tested in accordance with Method 501.1, Procedure 1 of MIL-STD-810. The storage temperature shall be 70°C, maintained for two hours and the operating temperature shall be 65°C. Prior to, during and following the test, the equipment shall meet the requirements of 3.7.1.2.

4.8.2 Humidity. - The equipment shall be tested in accordance with Method 507.1, Procedure II of MIL-STD-810, with measurements made during the last five hour period of each cycle. Prior to, during and following the test, the equipment shall meet the requirements of 3.7.2.

4.8.3 Altitude. - The equipment shall be tested in accordance with Method 500.1, Procedure 1 of MIL-STD-810 except that the pressure in Step 2 shall first be raised to 50,000 feet above sea level. Prior to, during and following the test, the equipment shall meet the requirements of 3.7.3.

4.8.4 Leakage (Immersion). - The equipment shall be tested in accordance with Method 512.1, Procedure 1 of MIL-STD-810 and meet the requirements of 3.7.4 prior to and following the test.

4.8.5 Blast. - Mount the microphone under test on the carriage of the Simulated Gun Blast Producing Equipment in accordance with U.S. Navy Drawing SK-N-864 with the front edge of the element in the test plane and with its axis coincident with that of the explosion chamber. Subject the element to 30 rounds of blast at a peak pressure of 9.5 psi. Determine compliance with 3.7.5.



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4.8.6 Shock, drop. - The equipment, wrapped in an eight inch square piece of 8 to 12 ounce canvas, secured with rubber bands, shall be dropped twelve times from a height of six feet onto a concrete floor. Prior to and following the test, the equipment shall meet the requirements of 3.7.6.

4.8.7 Bounce, loose cargo. - The equipment, canvas wrapped as in 4.8.6, shall be tested in accordance with Method 514.2, Procedure XI, Part 2 of MIL-STD-810 except that the equipment shall lay at random on the test table and no reorientation shall occur at the half hour periods. Prior to and following the test, the equipment shall meet the requirements of 3.7.7.

4.8.8 Salt fog. - The equipment shall be subjected to the salt fog test specified in Method 509.1, Procedure 1 of MIL-STD-810. No corrosion products shall be deposited on the test item by the facility. Inability of the equipment to meet the requirements of 3.7.8 shall constitute failure of this test.

4.9 Inspection for dimensional interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.8. When listed dimension is not within specified or design limits, it shall be considered a major defect.

a. Mounting dimensions of microphone housing and overall size.

b. Mating dimensions of microphone contacts and cable terminals.

c. Mating dimensions of jack JJ-055.

4.10 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirement of Section 5.



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4.11 Rough handling test (preparation for delivery). - When rough handling test in accordance with MIL-P-116 is required by the contract, the operational test of 4.7.2.3 shall be conducted to determine freedom from malfunction caused by rough handling.

5. PACKAGING

5.1 Preservation. - Preservation shall be level A, B or Commercial as specified (See 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. - Each Microphone M-116/G shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Each Microphone M-116/G shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. - No preservative required.

5.1.1.4 Unit packing. - Each Microphone M-116/G shall be individually unit packed Method III as follows: Secure the cord to the microphone with twine conforming to T-T-871, 3 ply. Cushion each microphone by wrapping with cushioning material conforming to PPP-C-843, type II, class B or PPP-C-795 or PPP-C-1797. Place the cushioned item within a close-fitting paperboard box conforming to PPP-B-566, variety 2, process II, style II or PPP-B-676, type I, variety 2. Close the box in accordance with the appendix of the applicable specification.

5.1.2 Level B. - Cleaning, drying, preservative application, and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. - Items shall be given the best degree of protection employed by the supplier to afford protection against corrosion, deterioration, and damage during shipment. When unit pack quantities are not specified in the contract or order, the unit pack shall contain that quantity used as the supplier standard. However, the contents of the unit pack shall not exceed 50 pieces or 25 pounds. Items weighing over 10 pounds shall be individually packed.



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5.2 Packing. - Packing shall be level A, B or Commercial, as specified (See 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. - A quantity of Microphone M-116/G unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.1.1 Unitization. - Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.



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5.2.2 Level B. - A quantity of Microphone M-116/G unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be"bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. - Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Microphone M-116/G unit packed as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

5.2.3 Commercial packing. - The item shall be packed to meet carrier acceptance and safe delivery to destination at lowest rates in compliance with requirements of carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Military marking. - In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.



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5.3.2 Commercial marking. - In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall as a minimum be marked as follows: Noun nomenclature; National Stock Number, or part number when NSN is not given; Government contract or purchase order number; quantity; contractor's policy/ procedures; exterior containers shall also be marked with the appropriate address markings. Markings may be applied by any means which provide legibility.

6. NOTES

6.1 Intended use. - Microphone, Dynamic M-116( )/G, mounted in Mask M14A2, is intended for use with radio communications equipment (AN/VRC-12) when operated under high ambient noise conditions encountered in armored vehicles.

6.2 Ordering Data. - Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, level B or commercial preservation and packing (See Section 5).

c. When the rough handling test is required (See 4.11).

d. The number of First Article samples required (See 4.3.1).

6.3 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: M-116W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.



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6.5 Environmental. - Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or procuring activity for recommended disposability methods.

Custodian:

Army - EL

Preparing Activity

Army - EL

Project No. 5965 - A153



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Fort Monmouth, New Jersey 07703

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## SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in use of the specification in procurement of products for ultimate use by the Department of Defense. This s is provided for obtaining information on the use of this specification which will insure that suitable prod can be procured with a minimum amount of delay and at the least cost. Comments and the return of this f will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comr and suggestions submitted on this form do not constitute or imply authorization to waive any portion of referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-M-55198A(EL) MICROPHONE, DYNAMIC M-116( )/G

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A GIVE PARAGRAPH NUMBER AND WORDING.

   B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional p attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED
ESC-FM 1068-68

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